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19 January 2010
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18 May 2001

PERFORMANCE SPECIFICATION SHEET

# CABLE, FIBER OPTIC, ONE TUBE, BLOWN OPTICAL FIBER, STANDARD AND ENHANCED PERFORMANCE, CABLE CONFIGURATION TYPE 5 (TUBE), APPLICATION B (SHIPBOARD), CABLE CLASS SM AND MM (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

CLASSIFICATION:

Fiber optic cable configuration type (Tube): 5

Fiber cable class: MM (graded-index, glass core and glass cladding, multimode)

SM (dispersion-unshifted, glass core and glass cladding, single mode)

DESIGN AND CONSTRUCTION:

Blown optical fiber tube:

Dimensions and configuration: See figure 1.

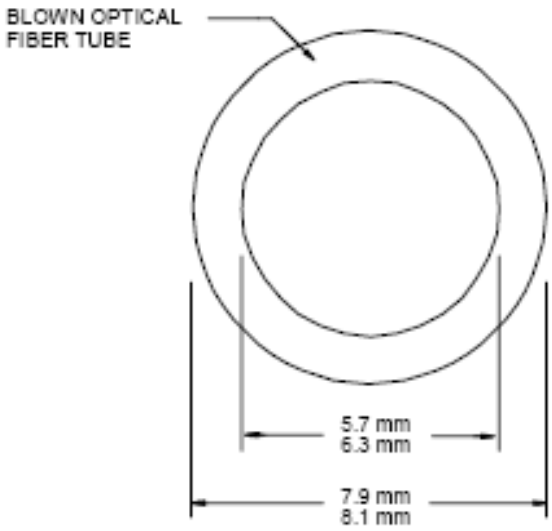



FIGURE 1. Blown optical fiber tube.

Short-term minimum bend diameter: 127 millimeters

Long-term minimum bend diameter: 127 millimeters



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Tensile loading: ≥89 Newtons

Tube material: The tube shall be composed of a low halogen, low smoke, and low toxicity polymer material.

Finished cable:

Dimensions and configuration: See figure 2. An outer jacket shall be applied over strength members and a central tube. The minimum outer jacket thickness shall be not less than 1.15 millimeters.

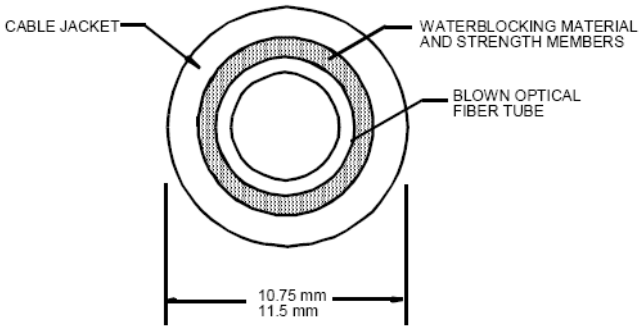



FIGURE 2. Single-tube cable.

Concentricity: ≥0.65

Jacket material: The overall jacket shall be composed of a low halogen, low smoke, and low toxicity polymer material.

Cable jacket color: Shall be black or blue

Mass per unit length: ≤120.0 kg/km

Short-term minimum bend diameter: 127 millimeters. (The short-term minimum bend diameter is to be used in all environmental and mechanical tests that specify a cable minimum bend diameter.)

Long-term minimum bend diameter: 127 millimeters

PERFORMANCE REQUIREMENTS:

Specimen lengths: 3 units, 0.305 kilometers each

Optical properties:

Attenuation rate: Not applicable

Change in optical transmittance: Not applicable

Crosstalk: Not applicable

Mechanical properties:

Tensile loading and elongation: Applicable, tensile loading ≥445 Newtons. Change in optical transmittance is not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube after the test.

Operating tensile loading: Applicable, tensile loading ≥267 Newtons. Change in optical transmittance is not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube during and after the test.

Dynamic bend: Not applicable



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Low temperature flexibility: Applicable, except the change in optical transmittance is not applicable. For standard performance, the exposure temperature shall be the minimum operating temperature. For enhanced performance, the exposure temperature shall be –40 °C. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube after the test.

Cyclic flexing: 500 cycles at +25±2 °C and 100 cycles at –28±2 °C. Change in optical transmittance is not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube after every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. The cycling may be halted to perform the ball bearing test.

Crush: Applicable, except that the load shall be 950 Newtons and the change in optical transmittance and crosstalk are not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube after the load is removed.

Cable twist bending: Not applicable.

Radial compression: Applicable, except the change in optical transmittance is not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube during and after the test.

Impact: Applicable, except that the drop hammer mass shall be 1.5 kilograms. Fifty cycles shall be conducted at +25±2 °C and 20 cycles shall be conducted at -40±2 °C. The change in optical transmittance is not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube after the test.

Corner bend: Applicable, except the change in optical transmittance is not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through each tube after the test.

Cable jacket tear strength: Applicable, except the cable jacket tear strength shall be not less than 35 Newtons per centimeter (N/cm) for standard performance cable.

Tube tensile strength and elongation: Tube specimens shall be tested in accordance with FED-STD-228, Method 3021 and 3031, with 2.5 cm (0.98 inch) benchmarks, 6.35 cm (2.5 inch) jaw separation, and a rate of travel of 25 cm/min (9.8 inch/min). The tensile strength of the tube shall be not less than 900 $N/cm^2$. The percent elongation-at-break shall be not less than 125 percent. Capstan grips may be used.

Hosing:

Low pressure: Applicable. Tube ends shall be capped with end caps during this test.

Hydrostatic: Not applicable

Dripping: Applicable

Cable scraping resistance: 750 cycles

Cable to cable abrasion: 500 cycles

Cable shrinkage: Applicable, except that the total shrinkage shall be not greater than 35 millimeters.

Pressure withstand: One end of the inner tube shall be capped and a static pressure of 1.4 MPa (200 psi) applied internal to the tube for 10 minutes. After the test, tubes shall show no evidence of splitting, cracking, or rupture.

Environmental properties:

Temperature range:

Operating: -28 to +65 °C

Nonoperating: -40 to +70 °C

Storage: -40 to +70 °C



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Temperature cycling: Change in optical transmittance is not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube at the high temperature plateau, the low temperature plateau, and after the test.

Thermal shock: Not applicable

Temperature humidity cycling: Change in optical transmittance is not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube at the high temperature plateau, the low temperature plateau, and after the test.

Storage temperature: Applicable, except the change in optical transmittance is not applicable. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube after the test.

Life aging: Applicable, except the change in optical transmittance is not applicable and a minimum test sample length of 150 meters may be used. For standard performance product, the test shall be conducted as specified in the basic specification. For enhanced performance product, the test shall be conducted as specified in the basic specification except that the jacket material shall be tested at +175 °C for 4 hours. A minimum of 2 meters of the sample shall be maintained at the minimum bend diameter throughout the test. A ball bearing with a minimum outer diameter of 4.0 millimeters shall pass through the tube after the test.

Off reel testing: Not applicable

Weathering: Not applicable

Fluid immersion:

Standard performance: Exposure to automobile gasoline and tap water are not required and the test temperature for lubricating oil exposure shall be 73 to 77 °C. The tensile strength retention of the cable jacket material after exposure to hydraulic fluid shall be not less than 30 percent.

Enhanced performance: Exposure to automobile gasoline and tap water are not required and the following test temperatures shall be used for the fluids indicated: fuel oil (98 to 100 °C), turbine fuel (48 to 50 °C), and lubricating oil (98 to 100 °C).

Flame extinguishing: Applicable. Tube ends shall be plugged with a non-flammable sealant to simulate end caps.

Smoke generation and flame propagation: Applicable. Tube ends shall be plugged with a non-flammable sealant to simulate end caps. The pass/fail criteria shall be as follows. The peak optical density and the average optical density of smoke produced shall be not greater than 1.6 and 0.55, respectively. In addition, the flame spread-time product at the 10-minute point shall be not greater than 27.5 meters-minutes when calculated in accordance with ASTM E84.

Shock: Applicable

Paint susceptibility: Applicable

Chemical properties:

Halogen content: <0.2 percent

Cross-link verification: This test is applicable for cables with cross-linked jackets only. The test shall be conducted in accordance with ICEA standard T-28-562 and run at 200 °C. The test shall be sequenced after the fluid immersion test in the qualification test sequence and in the group C conformance test sequence. The hot creep shall not exceed 100 percent and the hot creep set shall not exceed 10 percent.

Part or identifying number (PIN):

M85045/26-01S (Standard performance)

M85045/26-01E (Enhanced performance, Cross-linked outer jacket)



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Qualification by similarity:

Manufacturers who produce enhanced performance products for both MIL-PRF-85045/25 and this specification sheet, and are qualified under MIL-PRF-85045/25, and whose enhanced performance product passes the visual and mechanical, low temperature flexibility (cold bend), cyclic flexing, crush, impact, life aging (complete cable only), tensile loading and elongation, operating tensile load, low pressure hosing, radial compression, corner bend, cable abrasion resistance, smoke generation and flame propagation, and flame extinguishing inspections specified herein, are qualified under this specification sheet for enhanced performance product. This qualification by similarity is applicable if the same cable jacket and other materials are used in the previously qualified MIL-PRF-85045/25 cable and the enhanced performance cable under test.

Manufacturers who produce standard performance products for both MIL-PRF-85045/25 and this specification sheet, and are qualified under MIL-PRF-85045/25, and whose standard performance product passes the visual and mechanical, low temperature flexibility (cold bend), cyclic flexing, crush, impact, life aging (complete cable only), tensile loading and elongation, operating tensile load, low pressure hosing, radial compression, corner bend, cable abrasion resistance, smoke generation and flame propagation, and flame extinguishing inspections specified herein, are qualified under this specification sheet for standard performance product. This qualification by similarity is applicable if the same cable jacket and other materials are used in the previously qualified MIL-PRF-85045/25 cable and the standard performance cable under test.

Qualification by similarity for change to outer cable jacket:

Manufacturers who produce products for MIL-PRF-85045/25 using one particular overall cable jacket compound and are qualified under MIL-PRF-85045/25 for that compound and pass visual and mechanical, low temperature flexibility, durability of identification, smoke generation and flame propagation, flame extinguishing, water absorption, acid gas generation, toxicity index, crosslink verification and all applicable requirements indicated below for cables with a modified overall cable jacket compound are also qualified under MIL-PRF-85045/25 for cables with the modified overall cable jacket compound.

In addition to the requirements identified above, the following requirements are also applicable if a change is made in the outer cable jacket compound. If the base material of the modified cable jacket compound is not the same as the already qualified cable jacket compound, then the modified cable jacket shall also pass fluid immersion, paint susceptibility, jacket self adhesion or blocking, dripping, cable scraping resistance, and cable-to-cable abrasion. If the modified cable jacket compound results in process changes, then the cable with the modified jacket shall also pass hosing: low pressure, and cable shrinkage. If the modified jacket compound results in a color change then the modified jacket shall pass fungus resistance and halogen content. The modified jacket compound shall pass fungus resistance unless colorants are fungus inert in accordance with MIL-HDBK-454 and compliance to the halogen content requirement may be achieved by providing formulation of colorants. If the modified cable jacket compound results in a change in jacket color and the jacket material tensile strength and elongation results for the qualified cable jacket are not the same as those for the modified cable jacket, then the modified cable shall pass cyclic flexing, barometric pressure, life aging, and cable jacket tear strength.

Intended use: This cable is intended for use as a local cable and is not intended for installation in the cableways.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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Review Activities:
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Navy – EC, YD
Air Force – 02, 19, 99
DLA – CC

Preparing Activity:
Navy – SH
(Project 6015-2009-003)

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